Progressive exercises

E1 · Direct application. A code is a sequence of four digits chosen from 0, 1, . . . , 9. The repetitions and the digit 0 in first position are allowed. How many codes contain at least one 0?

E2 · Application and justification. Ten athletes run a race with no ties. How many different podiums, with a first, second and third place, can be obtained?

E3 · Method to organise. From seven distinct people, a group of three people is formed containing a person designated in advance. How many groups can be formed?

E4 · Choice of strategy. On a grid, we start from $(0;0)$ to reach $(4;3)$ by performing exactly four moves of one unit to the right and three upwards. How many different paths can be followed?

E5 · Synthesis or proof. A class has eight girls and seven boys. How many groups of four students containing exactly two girls can be formed?